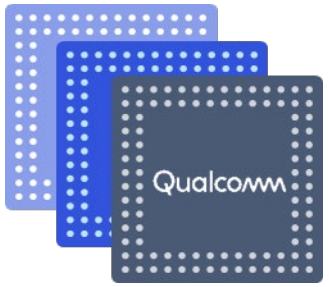
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QCM6490/QCS6490 and QCM5430/QCS5430 Camera Overview

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80-PT831-82 Rev. AC



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Revision History

Revision	Date	Description
AA	January 2021	Initial release
AB	April 2023	Editorial updates
AC	May 2023	Updated document title. Updated maximum real-time sensor input resolution and ZSL performance.

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Contents

- Spectra 570L Overview
- Image Front End
- Bayer Processing Segment
- Image Processing Engine
- References
- Questions?

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Spectra 570L Overview

Feature	QCM6490/QCS6490	QCM5430/QCS5430
Maximum real-time sensor input resolution (4:3 aspect ratio)	3x IFE: 22M@30fps 2x IFE-Lite: Mono/YUV interface for miscellaneous CV use cases	2 x IFE: 22M@30fps 2 x IFE-Lite: Mono/YUV Interface for miscellaneous CV use cases
Zero shutter lag (ZSL) performance	▪ 3x 22MP@30fps – Triple Camera ▪ 36MP+22MP@30fps – Dual Camera ▪ 36MP@30fps – Single Camera ▪ 64MP – Single Camera QCFA CSID Binning	▪ 2x 22MP@30fps – Dual Camera ▪ 36MP@30fps – Single Camera ▪ 64MP – Single Camera QCFA CSID Binning
Maximum non-ZSL Snapshot resolution (4:3 aspect ratio)	192MP (w/o MFNR) 48MP (MFNR)	
Camera interfaces	5 PHYs DPHY 1.2: 4/4/4/4/4 (2.5 Gbps/lane) CPHY 1.2: 4.5Gsps/T, 30.78Gbps 5x concurrent (3 Bayer, 2 YUV/Mono)	4 PHYs DPHY 1.2: 4/4/4/4 (2.5 Gbps/lane) CPHY 1.2: 4.5Gsps/T, 30.78Gbps 4x concurrent (2 Bayer, 2 YUV/Mono)
Video	Single Camera: 4K@30fps 10 bit / 1080p@120 / 720P@240 Refer to *QCM6490/QCS6490 and QCM5430/QCS5430 Linux Android Video Technical Overview* (80-PT831-89)	

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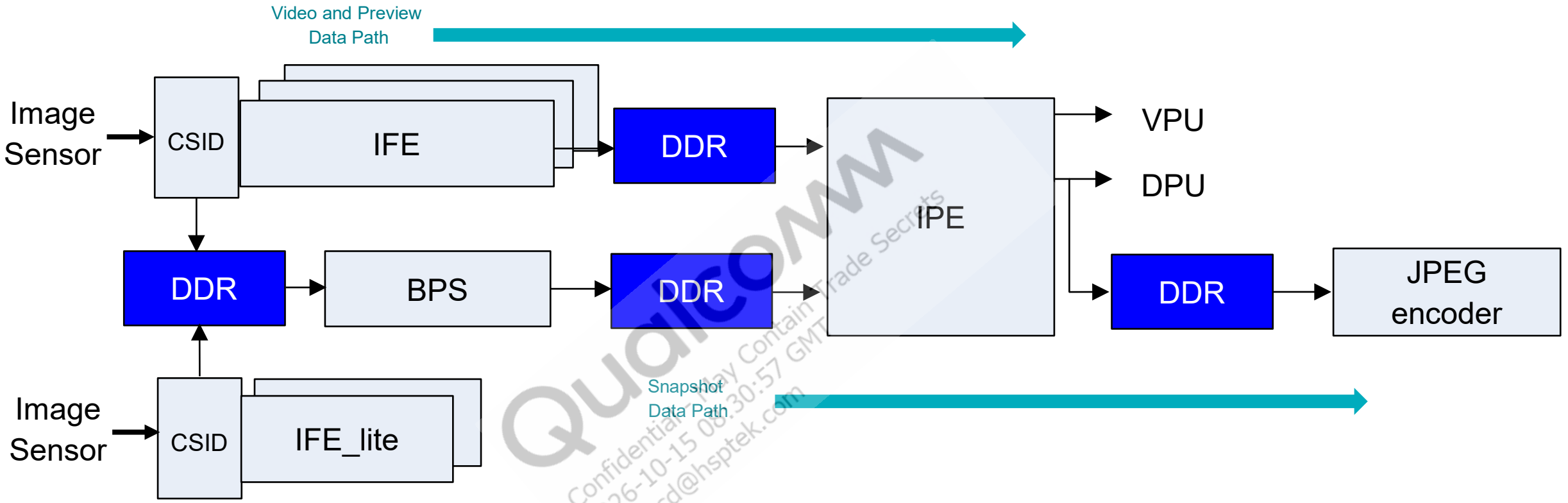
Spectra 570L Overview (cont.)

Feature	Spectra 570L
Outputs from BPS	YUV 4:2:0 Bayer ARGB
IQ and imaging improvements	▪ High-quality directional upscaler ▪ 2PD hardware support ▪ EIS processing ▪ Bayer and QCFA in-ISP binning (ZSL support for high-resolution sensors) ▪ HDR10 support ▪ Face detection w/ Deep Learning v2.0 ▪ Hardware remosaic 2x2 Bayer, 4x4 QCFA

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High-level Spectra 570L Architecture Overview



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- CSID = Camera Serial Interface Decoder module
- IFE (x3) = Image Front End
- IFE_lite (x2)
- BPS = Bayer processing segment (for Snapshot)

- IPE = Image Processing Engine
- VPU = Video Processing Unit (CODEC)
- DPU = Display Processing Unit
- Two IFE (IFE0/IFE1) have HVX streamer interfaces only

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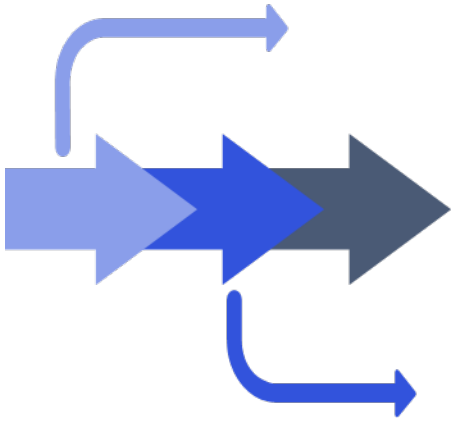
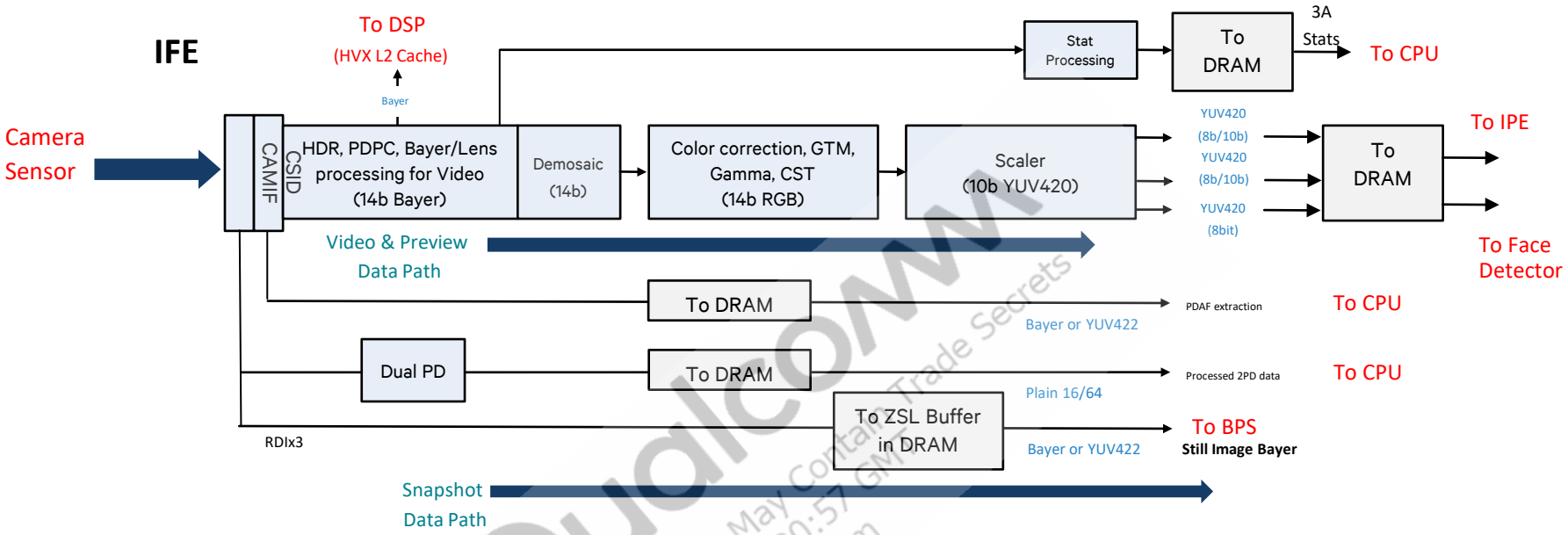


Image Front End

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High-level Architecture Overview of Spectra 570L: IFE

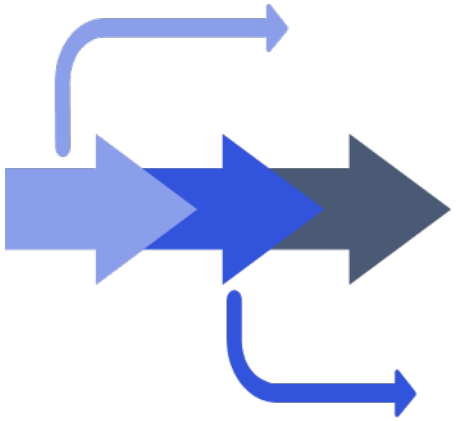


- **IFE x 3**
  - Bayer processing for Video/Preview
  - Stats for 3A (Bayer grid, histogram/Bayer focus, etc.)
  - 4 tap-in/out points for HVX streaming
  - Dual PDAF
  - Separate Preview and Video output
  - Triple camera support
  - Only IFE0 and IFE1 support HVX tapping

- **CSID**
  - CSID Decoder
  - 3 RDI (Raw Data Interface) output to DDR
  - IFE_lite is separate from IFE

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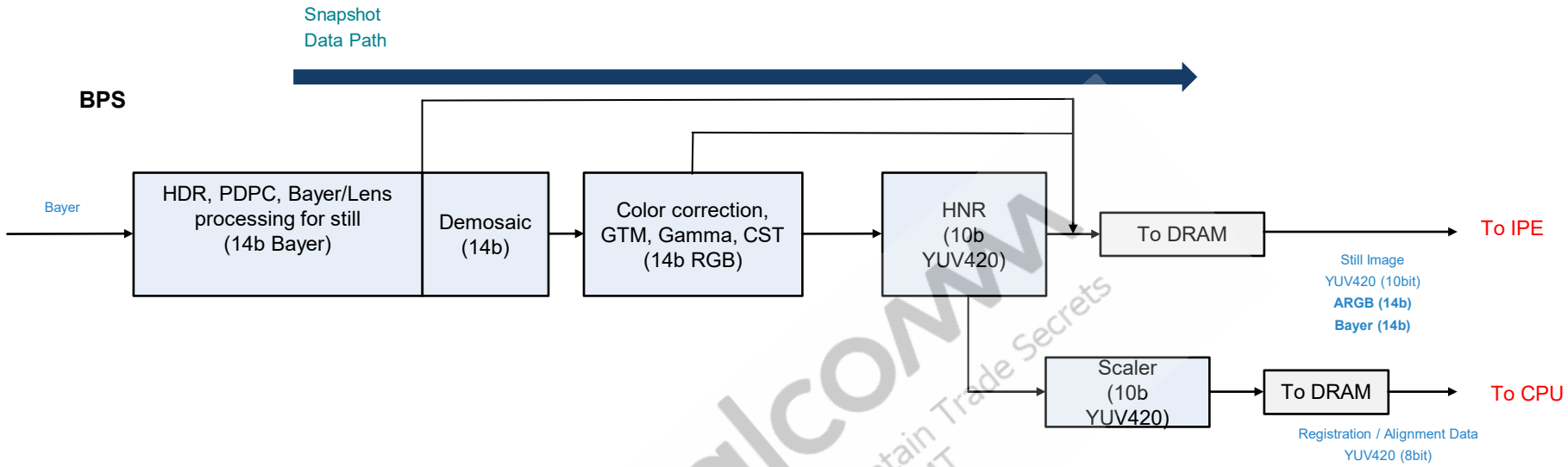
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Bayer Processing Segment

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High-level Architecture Overview of Spectra 570L: BPS

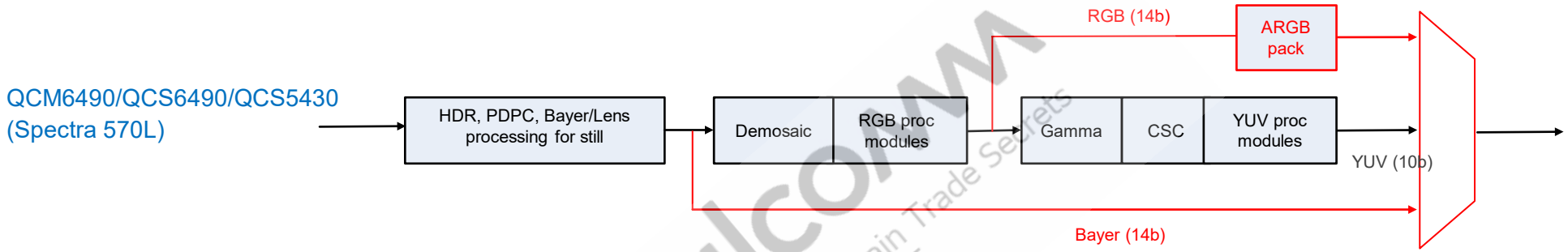


- BPS
  - Snapshot path only
  - Bayer processing, e.g., demosaic, PD pixel correction, lens shading correction, etc.
  - Hybrid noise reduction (HNR) for spatial and frequency domain
  - Downscaling for registration and alignment of sequential frames for multi-frame processing
  - BPS can perform noise reduction for YUV sensors (not a common use case)
  - Multiple tap-out points

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Spectra 570L BPS: Multiple Tap-Out Points



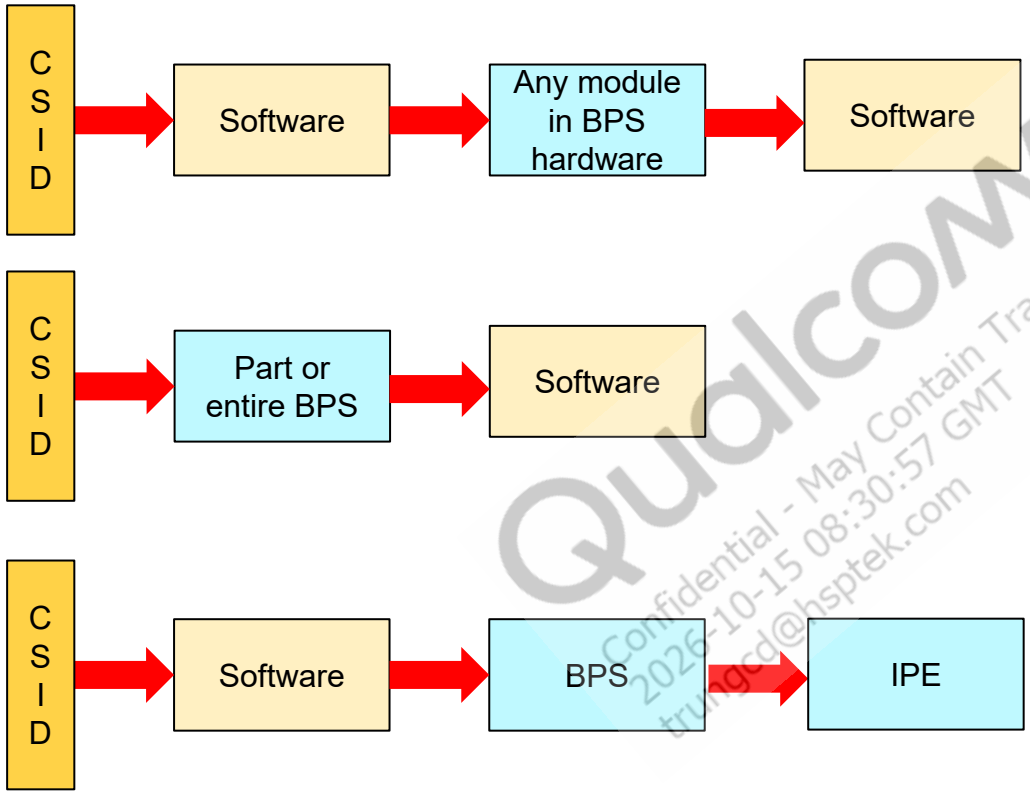
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- Add Bayer tap-out point before demosaic module
- Add RGB tap-out point before gamma module

Input	Output
Bayer	▪ Bayer ▪ RGB ▪ YUV
YUV	YUV

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Spectra 570L BPS: Hybrid Hardware and Software ISP by BPS



Spectra 570L

Input	Output
Bayer	▪ Bayer ▪ RGB ▪ YUV
YUV	YUV

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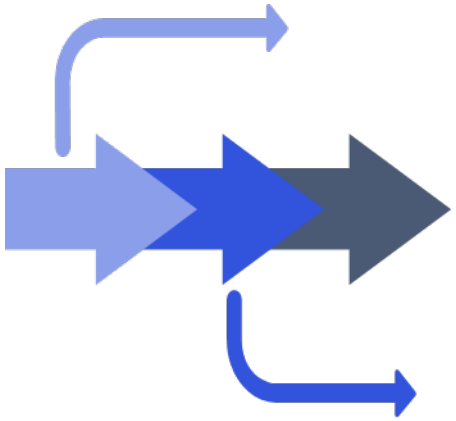
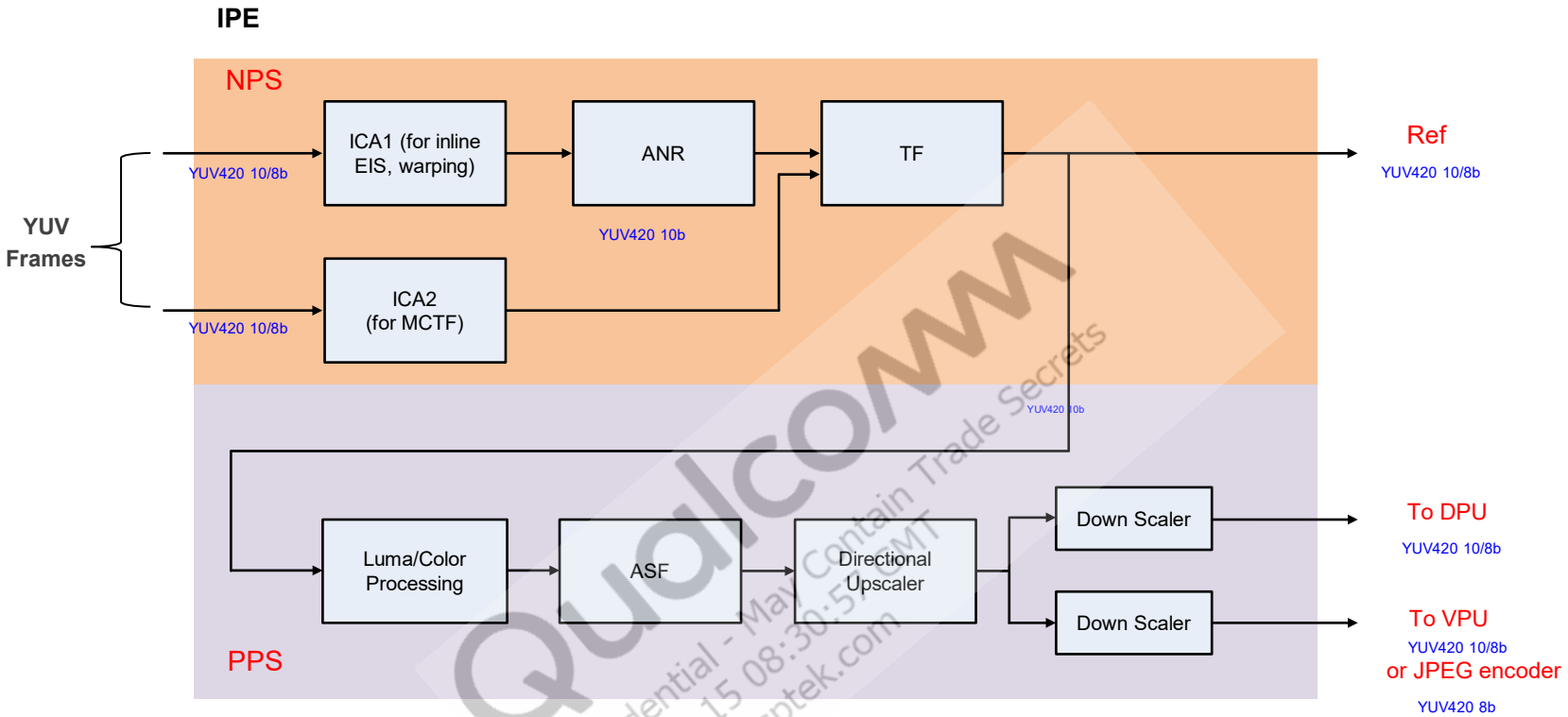


Image Processing Engine

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High-level Architecture Overview of Spectra 570L: IPE



- IPE x 1
  - NPS (ICA/ANR/TF) + PPS (CAC/LTM/ASF and color processing)

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Face Detection

	Spectra 570L
Implementation	▪ Software DL FD 2.0 ▪ Hexagon DSP ▪ Full deep-learning implementation
Accuracy*	> 95%*
Improvements	▪ Challenging Backlit faces ▪ Masked, partial faces ▪ Faces in low lighting ▪ Rotated faces ▪ Low power consumption ▪ Low DSP utilization

* Qualcomm Technologies, Inc. (QTI) internal faces database.

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Android Multicamera API

- Multicamera API support is mandatory for Android R
  - Requires multicamera to pass Compatibility Test Suite (CTS)
    - Passing CTS enables third party applications to access multicamera solutions
    - CTS pass level (LIMTED/FULL/L3) is still determined by OEM
  - ITS compliance is mandatory

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Pros and Cons

- Pros
  - Third party camera applications are heavily used by end users
    - Passing CTS enables more features for third party applications
  - Same logical camera ID for Spatial Alignment and Transform (SAT)/Bokeh modes
  - Allows applications to access YUV/RAW output of individual physical cameras
    - Easier to run custom algorithms in application
  - New API increases ability for third party applications to use features, i.e., Bokeh
  - Easier to improve certain key performance indicators (KPIs) (dependent on hardware capability)
    - Better SAT ←→ Bokeh switching time
  - Reduced effort for OEMs to support Google CameraX requirements
- Cons
  - More CTS test cases introduced
  - Newer APIs for application developers

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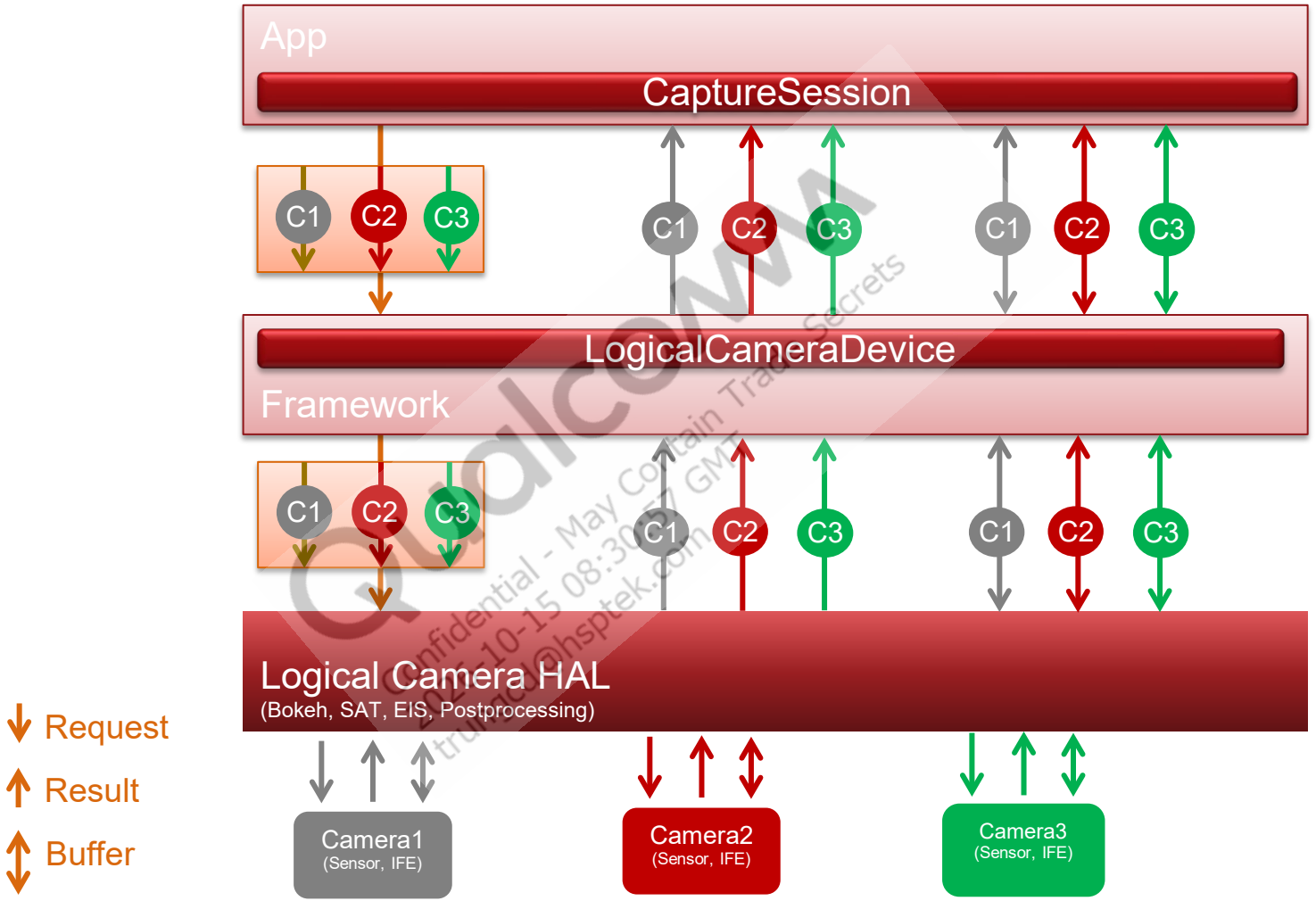
High-level Software Changes

- A new multicamera use case will be built to fully support the new API
  - New implementation needed to fit complex requirements, i.e., SAT, Bokeh, multicamera API
- Snapshot features
  - All Feature2 work from other CamX-based chipsets can be fully leveraged and reused

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Android Multicamera API Call Flow

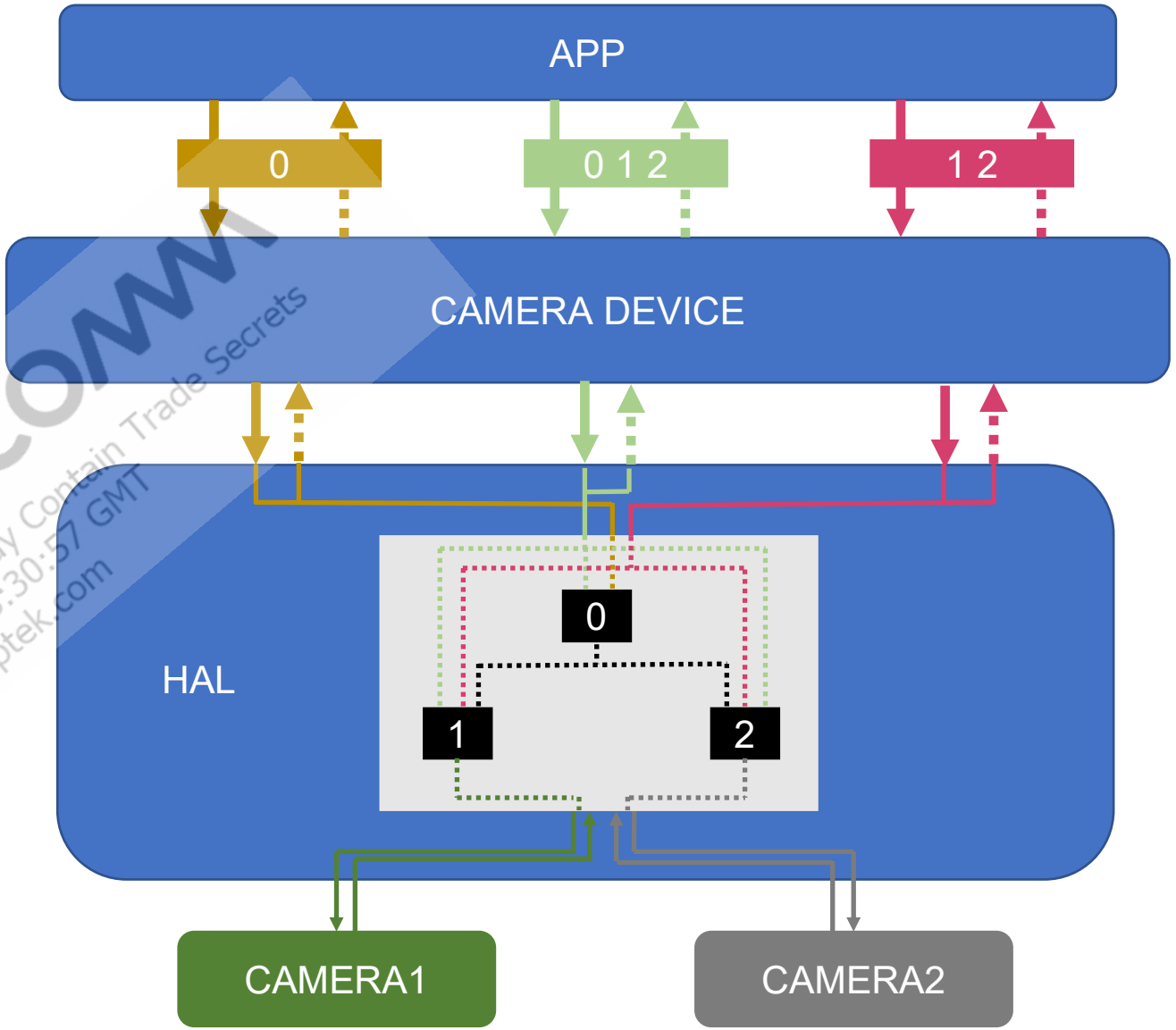


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Android Multicamera API

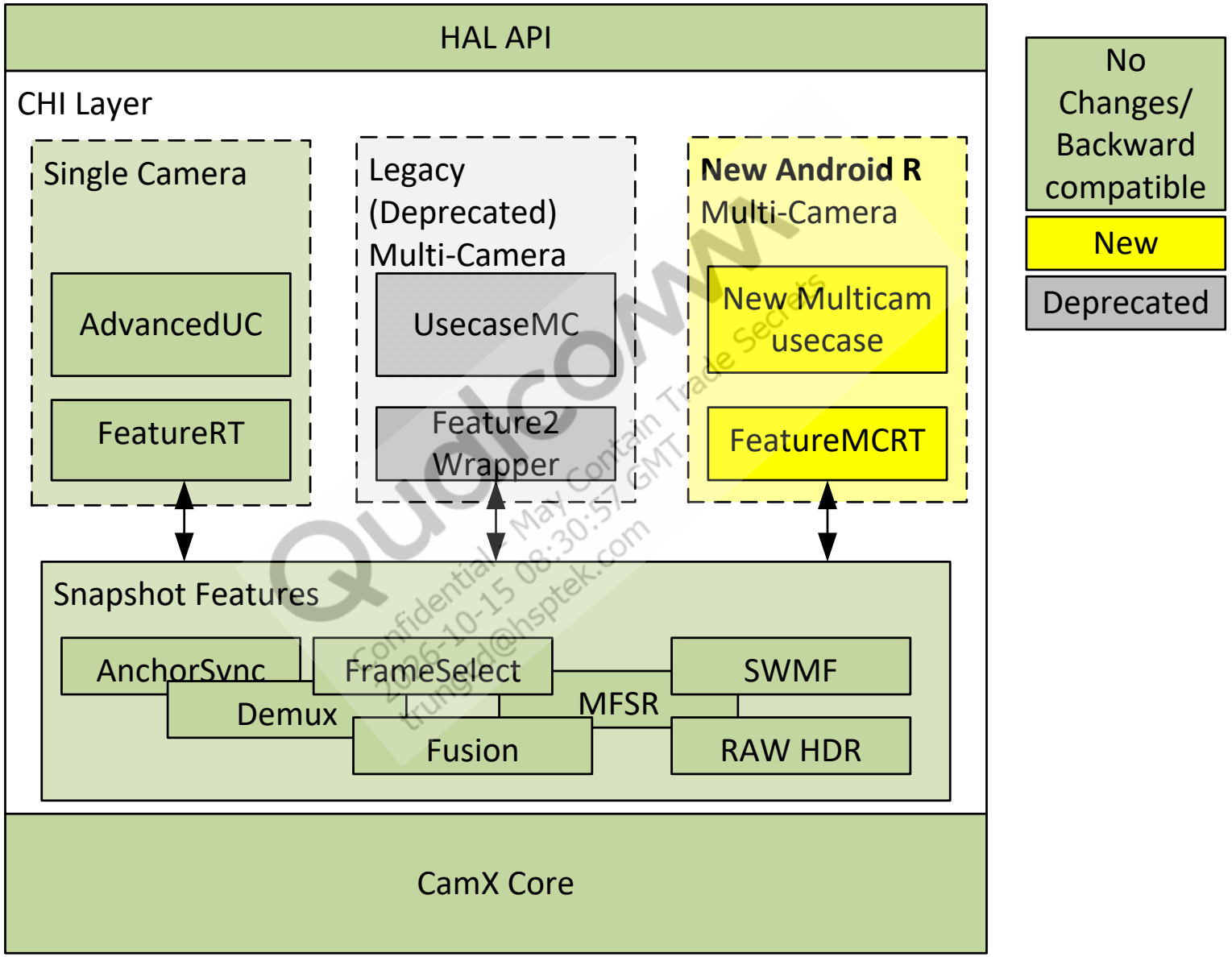
- Logical request splitting
  - Requests to each camera (logical or physical) are handled separately
  - In the block diagram, three requests are submitted:
    - To logical camera ID 0
    - To logical camera ID 0, physical cameras ID 1 and ID 2
    - To physical cameras ID 1 and ID 2
  - The same physical cameras can be accessed independently or in a combination



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High-level Software Architecture



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Why a New Use Case?

- usecasemc is not CTS or ITS compliant
  - CTS compliance requires special hacks in usecasemc but may not benefit third party applications
  - YUV Snapshot reprocessing not supported in usecasemc (CTS requirement as well)
- Bokeh and SAT require separate logical cameras in existing usecasemc
- No support for non-ZSL or quadra color filter array (QCFA) snapshots in usecasemc
- No support for depth cameras in usecasemc
- usecasemc is not fully Feature2 compliant
  - Requires custom hacks to enable advanced frame selection logic
- usecasemc does not support physical camera streams
  - Metadata not supported for physical cameras in existing usecasemc
- Very restricted support for stream combinations in usecasemc
  - Adding new streams has high impact on code
- Future Android enhancements will be hard to implement on existing use cases
  - Implementation on existing use cases will destabilize all existing usecasemc
  - It will not be possible for customers to leverage any existing CamX-based chipset's code if we try to enhance usecasemc

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New Multicamera Use Case Feature Support Matrix

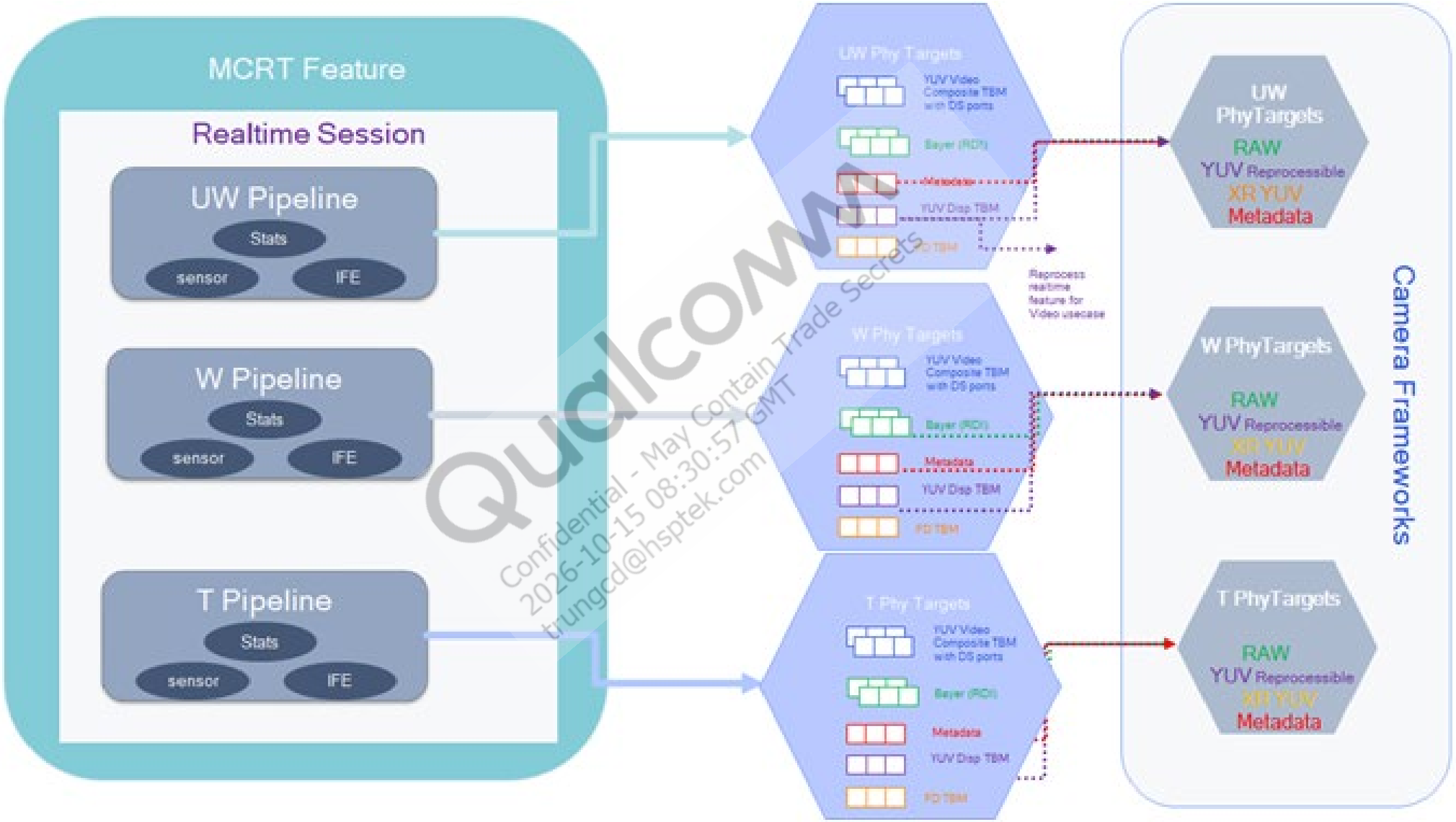
- Only major features listed (not a complete list)

Feature	New multicamera use case
SAT	Yes
Bokeh	Yes
Logical camera Preview/Video YUV callbacks	Yes
Logical camera Snapshot YUV callbacks	Yes
RAW callbacks	Yes
Feature2 Snapshot support	Yes
Physical camera Preview/Video YUV callbacks	Yes (based on hardware capability)
Physical camera Snapshot YUV callbacks	Yes (based on hardware capability)
Sensor-sized JPEG callbacks	Yes (based on hardware capability)
Depth/ToF camera support	Yes
Non-ZSL Snapshot support	Yes (without Preview)
QCFA full-size Snapshot support	Yes (without Preview)
Live Preview during multiframe Snapshot	Yes
Improved KPIs for SAT ←→ Bokeh switching	Yes
YUV Reprocessing support	Yes
CTS Compliance	Yes
ToF support for Bokeh	Yes
Fallback based on scene/thermal/sensor limits	Yes

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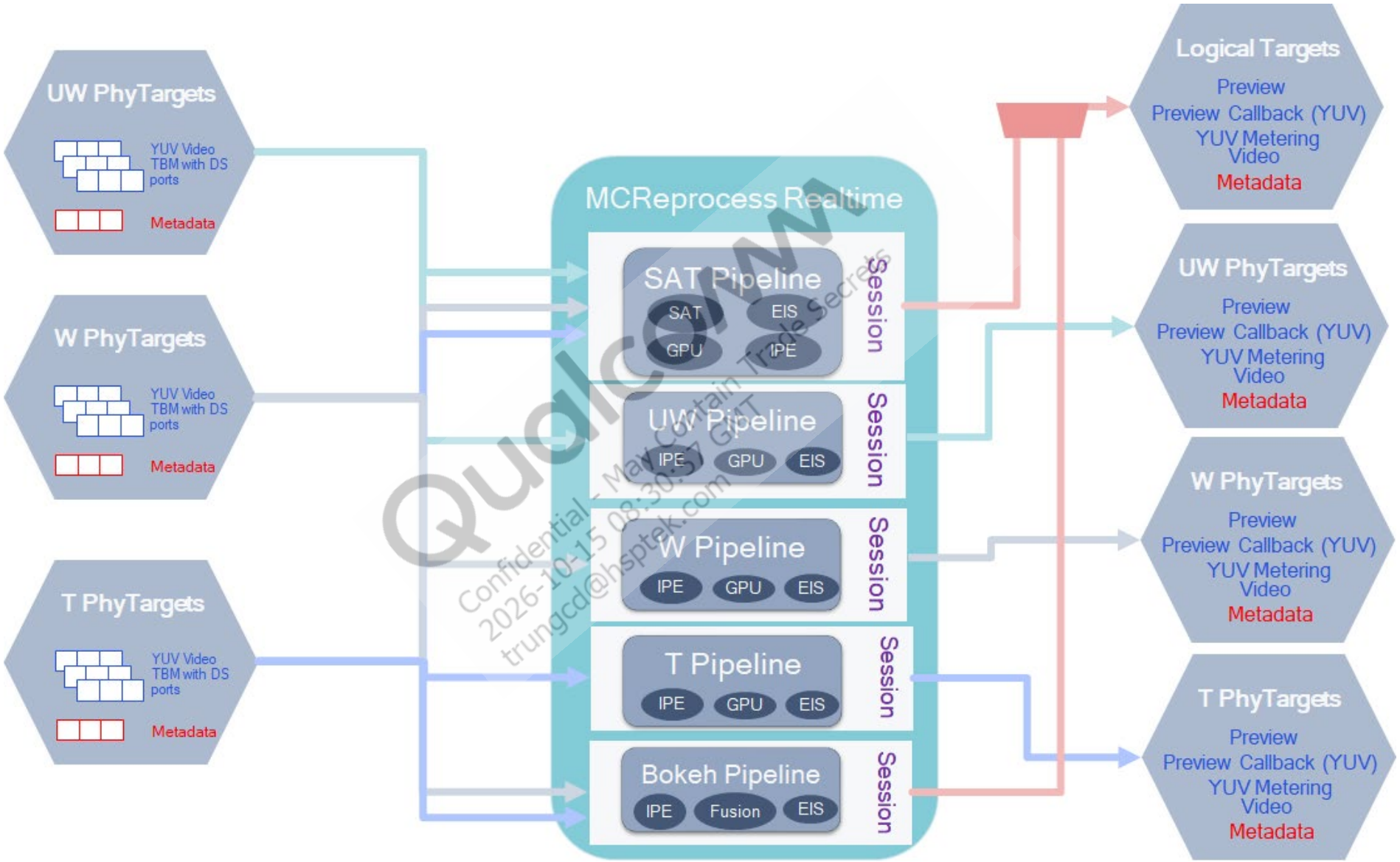
Realtime Feature Example



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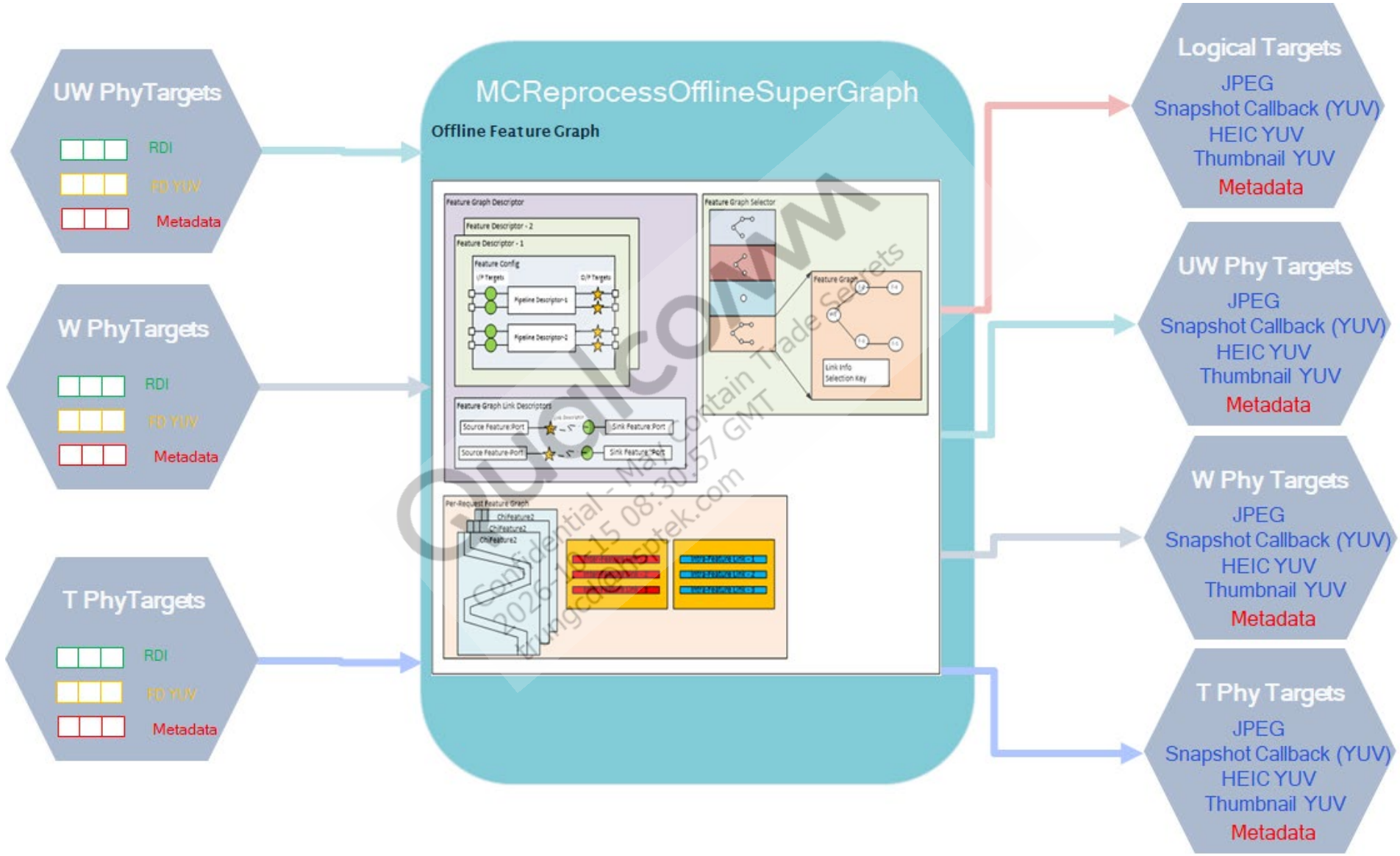
Offline Preview/Video Feature Example



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Snapshots



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References

Documents	
Title	**Number**
Qualcomm Technologies, Inc.	
Qualcomm Spectra Camera ISP Tuning guide	80-NK872-13
Linux Camera Debugging Guide	80-P9301-61
Chi API Specifications for Qualcomm Spectra 2xx Camera	80-PC212-1
Creating a Project in Chromatix 7	VD80-P9301-201
Chromatix 7 Mode Tree: Inheritance, Use Cases, Scenes, and Other Scenarios (Part 1)	VD80-P9301-202A
Chromatix 7 Mode Tree: Inheritance, Use Cases, Scenes, and Other Scenarios (Part 2)	VD80-P9301-202B
Chromatix 7 ISP Tuning Basics (Part 1)	VD80-P9301-203A
Chromatix 7 ISP Tuning Basics (Part 2)	VD80-P9301-203B
Chromatix 7 ISP Tuning Basics (Part 3)	VD80-P9301-203C
Camera 3A Tuning Guide	80-NK872-14
Introduction to Camera 3A v7.0	80-P9301-41
Qualcomm 3A 7.0 Deep Dive: AWB	80-P9301-99
Qualcomm 3A 7.0 Deep Dive: AEC	80-P9301-100
Qualcomm Spectra 2xx Deep Dive	80-P9301-60

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References (cont.)

Documents	
Title	**Number**
Qualcomm Spectra 2xx ISP Overview	VD80-P9301-14
Qualcomm Spectra 2xx Deep Dive: Pipeline introduction	VD80-P9301-60A
Qualcomm Spectra 2xx Deep Dive: HNR	VD80-P9301-60B
Qualcomm Spectra 2xx Deep Dive: ASF	VD80-P9301-60C
Qualcomm Spectra 2xx Deep Dive: TF	VD80-P9301-60D
Qualcomm Spectra 2xx Deep Dive: ICA	VD80-P9301-60E
Qualcomm Spectra 2xx Deep Dive: ANR	VD80-P9301-60F
Qualcomm Spectra 2xx Deep Dive: 2D LUT	VD80-P9301-60G
Qualcomm Spectra 2xx Deep Dive: Chroma Suppression	VD80-P9301-60H
Qualcomm Spectra 2xx Deep Dive: GRA	VD80-P9301-60I
Qualcomm Spectra 2xx Deep Dive: LTM	VD80-P9301-60J
Qualcomm Spectra 2xx Deep Dive: GIC/PNR	VD80-P9301-60K
Qualcomm Spectra 2xx Deep Dive: ABF	VD80-P9301-60L
Qualcomm Spectra 2xx Deep Dive: BPC/BCC, PDPC	VD80-P9301-60M

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References (cont.)

Acronyms	
Acronym or term	**Definition**
BPS	Bayer processing segment
CPP	Camera post processor
CSID	Camera Serial Interface Decoder
CTS	Compatibility Test Suite
CVP	Computer Vision Processor
FD	Face detection
DPU	Display Processing Unit
EIS	Electronic image stabilization
EVA	Engine for Visual Analytics
HAL	Hardware abstraction layer
HNR	Hybrid noise reduction
HDR	High Dynamic Range
ICA	Image Correction and Adjustment
IFE	Image front-end engine
IPE	Image processing engine
ISP	Image Signal Processor
MCTF	Motion compensation temporal filtering

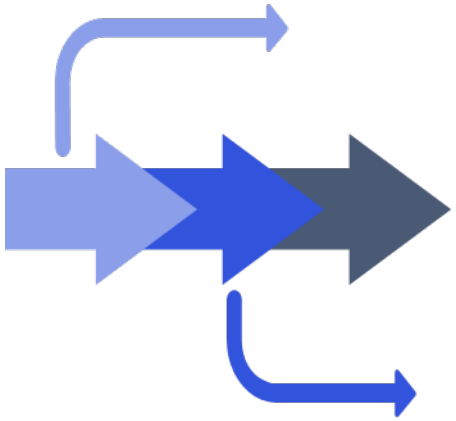
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References (cont.)

Acronyms	
Acronym or term	**Definition**
MFNR	Multiframe noise reduction
MFHDR	Multiframe HDR
PD	Phase difference
QCFA	Quadra color filter arrangement
QTI	Qualcomm Technologies, Inc.
RGB	Red Green Blue
SAT	Spatial Alignment and Transform
SLDC	Strong Lens Distortion Correction
QCFA	Quadra Color Filter Array
VFE	Video front end
VPE	Video processing unit
ZSL	Zero shutter lag

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Questions?

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